Severe diarrhea caused by cholera toxin-producing vibrio cholerae serogroup O75 infections acquired in the southeastern United States.

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BACKGROUND: From 2003 through 2007, Vibrio cholerae serogroup O75 strains possessing the cholera toxin gene were isolated from 6 patients with severe diarrhea, including 3 in Georgia, 2 in Alabama, and 1 in South Carolina. These reports represent the first identification of V. cholerae O75 as a cause of illness in the United States. V. cholerae O75 was isolated from a water sample collected from a pond in Louisiana in 2004. Subsequently, 3 V. cholerae isolates from Louisiana (2 from patients with diarrhea in 2000 and 1 from a water sample collected in 1978) that had been previously reported as serogroup O141 were also discovered to be serogroup O75. RESULTS: All 8 patients who were infected with V. cholerae O75 were adults who became ill after consuming seafood; 2 had eaten raw oysters traced back to the Gulf Coast of the United States. All 10 isolates possessed the cholera toxin gene and were susceptible to 10 antimicrobials. One clinical isolate and 1 environmental (water) isolate had the same pulsed-field gel electrophoresis pattern; 4 clinical isolates shared a common pulsed-field gel electrophoresis pattern. CONCLUSIONS: The occurrence of these cases over many years and the concurrent identification of V. cholerae O75 in water from a Gulf Coast state suggest that these strains may survive for long periods in this environment. The patients' exposure histories suggest that infection can be acquired from consumption of raw oysters from the Gulf Coast. Clinicians and public health authorities should be vigilant for the occurrence of new toxigenic serogroups of V. cholerae that are capable of causing severe diarrhea.